

What is MLOps?

Definition

MLOps (Machine Learning Operations) combines **Machine Learning, DevOps, and Data Engineering** to automate the development, deployment, monitoring, and maintenance of machine learning models.

Why is it Important?

A machine learning model with high accuracy is **not enough**. It must also be:

- Deployed for users
- Continuously monitored
- Updated with new data
- Maintained over time

MLOps automates these tasks, making models **production-ready**.

Key Points

ML + DevOps + Data Engineering

Automates the ML lifecycle

Improves collaboration

Ensures reliability and scalability

DevOps vs. MLOps

Aspect	Traditional DevOps	MLOps
Focus	Software Development	Machine Learning Development
Version Control	Code	Code + Data + Models
Testing	Software Tests (Unit, Integration, E2E)	Model Accuracy, Data Drift, Bias, Performance
Deployment Target	Applications & Services	ML Models as APIs, Embeddings, Edge Devices
Monitoring	Application Performance & Infrastructure Health	Application, Model Performance, Data Quality, Business Impact

Why MLOps is Needed?

Traditional ML focuses on **building models**, while MLOps ensures they work effectively in production.

Challenges Without MLOps

- Manual deployment
- No version control
- Model drift over time
- Manual retraining
- Difficult to scale

Benefits of MLOps

- Automated CI/CD deployment
- Version control for code, data, and models
- Continuous monitoring
- Automatic retraining
- Scalable cloud deployment

Example

A **fraud detection model** in banking is automatically monitored and retrained as fraud patterns change.

MLOps Lifecycle

Workflow

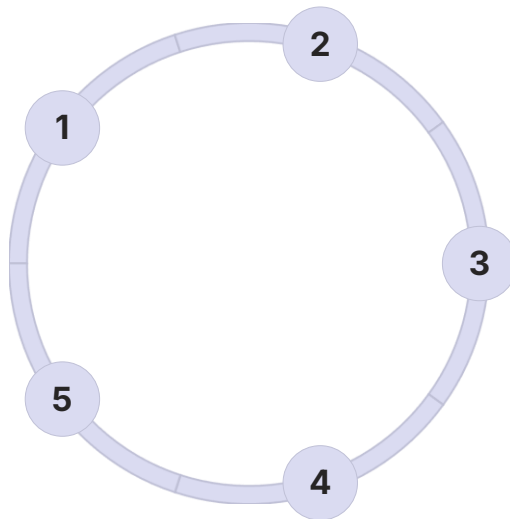
Steps

Data Collection

Gather data from databases, APIs, sensors, or cloud storage.

Monitoring & Retraining

Monitor performance, detect drift, and retrain when needed.



Data Preparation

Clean, transform, and prepare data for training.

Training & Evaluation

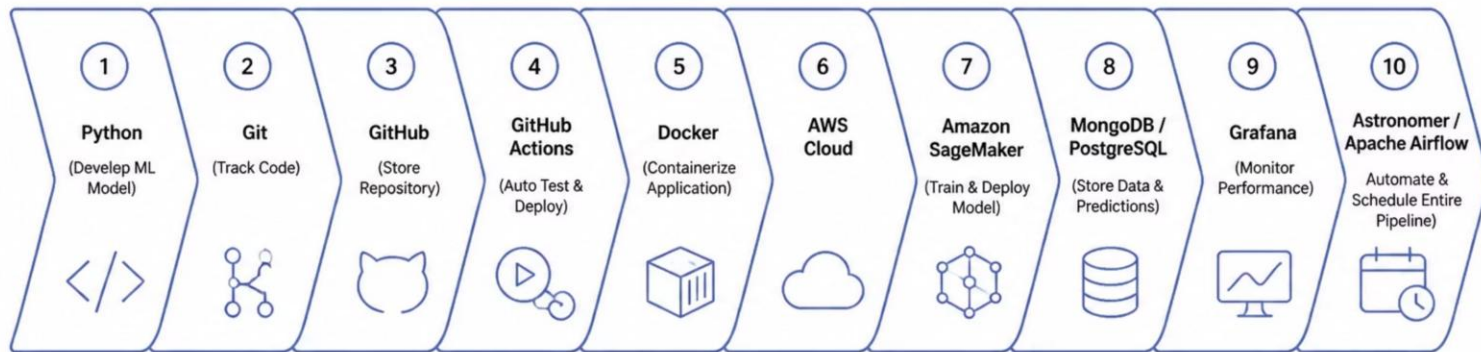
Train the model and evaluate its performance.

Deployment

Deploy the model as an API or cloud service.

MLOps Tools & Architecture

Toolchain



Real-World Applications

Applications



Healthcare

- Disease prediction
- Medical image analysis
- Patient monitoring



Banking

- Fraud detection
- Credit scoring
- Risk assessment



E-commerce

- Recommendation systems
- Customer behavior analysis
- Demand forecasting



Manufacturing

- Predictive maintenance
- Quality inspection
- Defect detection



Transportation

- Traffic prediction
- Route optimization
- Autonomous vehicles

"Building a machine learning model is only the beginning. The real challenge is deploying it, monitoring its performance, and continuously improving it in production. MLOps provides the tools and practices to make machine learning systems reliable, scalable, and ready for real-world applications."

MLOps Roadmap

